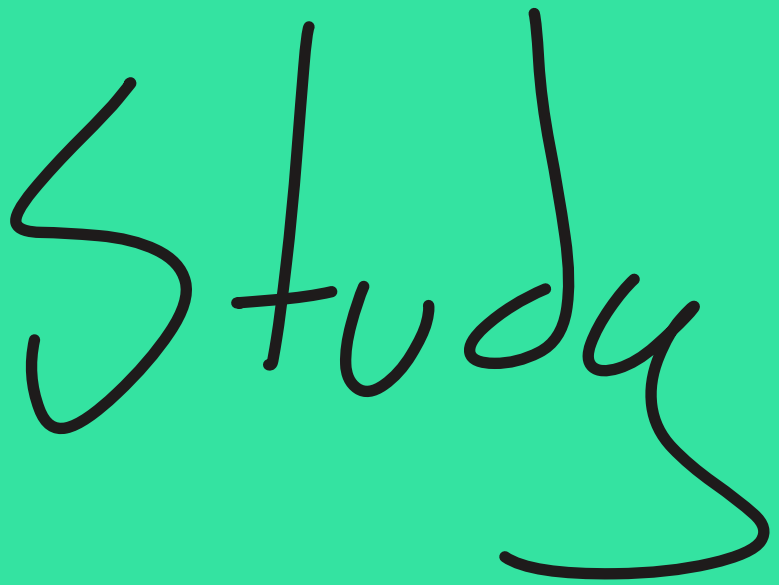

Study

Simplify

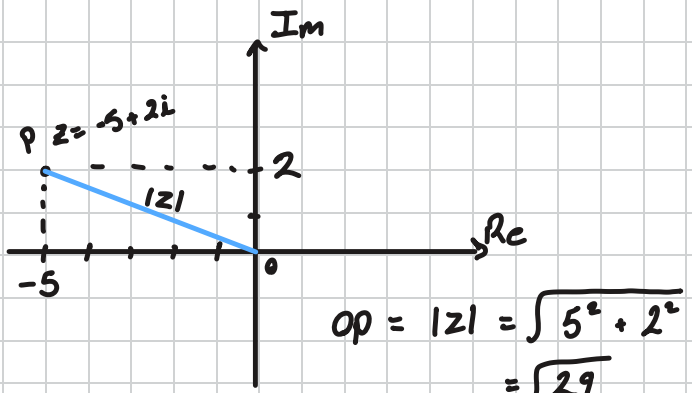
a) $\frac{2+3i}{4-i} \times \frac{4+i}{4+i} = \frac{(2+3i)(4+i)}{(4+i)(4-i)} = \frac{8-3+(2+12)i}{16+1} = \frac{5}{17} + \frac{14}{17}i$

b) $\frac{i}{1+i\sqrt{3}} = \frac{i(1-i\sqrt{3})}{(1+i\sqrt{3})(1-i\sqrt{3})} = \frac{\sqrt{3}+i}{1+3} = \frac{\sqrt{3}}{4} + \frac{1}{4}i$

① $z = -5 + 2i$

$Re(z) = -5$

$Im(z) = 2$



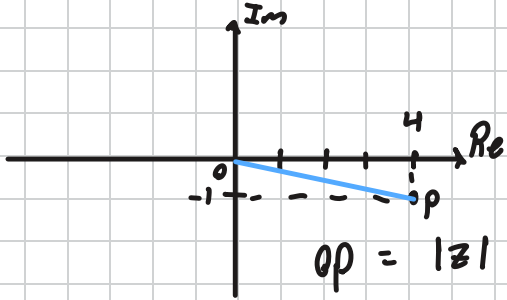
$OP = |z| = \sqrt{5^2 + 2^2}$

$= \sqrt{29}$

② $z = 4 - i$

$Re(z) = 4$

$Im(z) = -1$



$OP = |z| = \sqrt{4^2 + 1}$

$= \sqrt{17}$

$y = f(x)$ differentiable

$y' =$

Complex numbers

① $z = -5 + 2i$

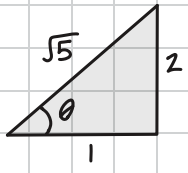
real: -5 imaginary: $2i$

② $z = 1 + 2i$

$r = |z| = \sqrt{a^2 + b^2}$

$= \sqrt{1^2 + 2^2}$

$= \sqrt{5}$



$\theta = \tan^{-1}\left(\frac{2}{1}\right)$

$= 63^\circ$

$z = a + bi = r\left(\frac{a}{r} + \frac{b}{r}i\right)$

$= r(\cos\theta + i\sin\theta)$ polar form

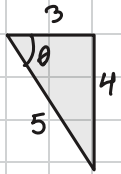
$= \sqrt{5}(\cos 63^\circ + i\sin 63^\circ)$

(3)

$z = 3 - 4i$

$r = \sqrt{3^2 + 4^2}$

$= 5$



$\tan^{-1}\left(\frac{-4}{5}\right) = -39°$ ← degrees

$\frac{-4}{5} = -0.8$ ← radians

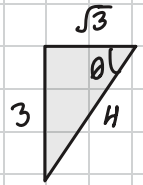
(4)

$z = -\sqrt{3} - 3i$

$r = \sqrt{3 + 9}$

$= \sqrt{12}$

$= 4$



$\theta = \frac{3}{\sqrt{3}}$ rad

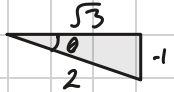
$\theta = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right)$

(5)

$z = \sqrt{3} - i$

$r = \sqrt{3 + 1}$

$= 2$



$\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \theta$

$\theta = -\frac{1}{\sqrt{3}}$ rads

